

LAB 3 – WRITTEN JUSTIFICATION

Component Modelling & Architectural Pattern Selection

1. Architecture Selection

For the Self-Service Coffee Kiosk System, we selected the Layered Architecture. Each module is responsible for some part of the functionality of the system and is organized into separate layers with particular responsibilities.

2. Architectural Choice – Why Layered Architecture?

First of all, the Layered Architecture provides good separation of responsibilities between different modules. Touchscreen can deal with customer interaction, whereas Order Manager is responsible for the logic of placing an order. Next, payment processing, receipt printing and information about menu and prices can be managed by appropriate components.

Second, the kiosk is a standalone system, thus it does not require many separately running services and constant deployment. Application of microservices architecture will result in redundant network communication and management of these services.

3. Security Advantage

It is particularly important due to the fact that the kiosk processes payments using credit cards. In the Layered Architecture, all payment functionality can be encapsulated in the Payment Service, thus no other component will process payments directly. The payment layer can use all necessary security measures like secure APIs, encryption, validation and other controls. Thus, less components will have access to the payment information.

4. Performance Advantage

As it was mentioned before, the kiosk has to complete an order in less than 60 seconds, thus it requires that the ordering process to be fast enough. In the Layered Architecture, it is achieved by keeping the order placing process simple. Customer chooses the type of coffee and its size on the touchscreen. Order Manager processes the order and contacts the necessary services for payment, menu and receipt printouts.

5. Conclusion

To summarize, Layered Architecture suits well the Self-Service Coffee Kiosk System. First of all, it is rather simple but takes into account the important requirements. Each component is assigned some responsibility, secure payment processing is provided and order placing process is made faster.